



Nilkamal School of Mathematics Applied Statistics & Analytics

DEPARTMENT OF STATISTICS

M.Sc. Statistics and Data Science

2025-2026

Beyond Black-Scholes:

An Event Study of Call Option Pricing in the Indian
Financial Landscape Using Taguchi Methodologies

Project Mentor

Pradnya Khandeparkar

Group number – 1

NAME	ROLL-NO	SAP-ID
Eashal Dbritto	<i>A014</i>	86062500061
Samiksha Kabra	<i>A023</i>	86062500017
Omkar Navle	<i>A037</i>	86062500050
Liz Rodrigues	<i>A050</i>	86062500051
Mohammed Aamir Vahedna	<i>A061</i>	86062500028

Acknowledgement

We, the students of NSOMASA, MSc Statistics and Data Science, Semester 1, acknowledge our gratitude to all the individuals associated with this project.

First of all, we would like to pay our respect to our mentor, Dr. Pradnya Khandeparkar, who has imparted us with knowledge, inspiration, and motivation throughout the course of the project. It would not have been possible for us to complete the same without her sincere and affectionate help.

Secondly, we would like to thank our RT professor, Mr. Vaibhav Vasundekar, for guiding us and keeping us motivated from the start till the end of this project. We would also like to extend our gratitude to our Dean, Dr. Sushil Kulkarni, for giving us an opportunity to work on such projects.

Next, we would like to thank SVKM's Library for allowing us to access the portal through which we got the required research papers, datasets, and study material needed for the project.

Lastly, we thank our faculty, parents, and friends who contributed to some extent to complete the same.

Declaration

We declare that the dissertation titled "Beyond Black-Scholes: An event study of call option pricing in the Indian Financial landscape using taguchi methodologies" is our original work carried out during the academic year 2025-2026 under the guidance of Dr. Pradnya Khandeparkar, Department of Mathematics, Applied Statistics & Analytics, NMIMS University. This work has not been submitted to any other institution for any degree or diploma. We have followed ethical research standards and properly referenced all external sources.

Abstract

This study investigates how the Taguchi L9 orthogonal array method can be combined with the Black-Scholes Model (BSM) to assess the pricing of European call options. The research examines the influence of four critical parameters: the underlying asset price (S_0), strike price (K), interest rate (r), and volatility (σ) on the value of a European call option at a specific time. By utilizing the Taguchi L9 design, the number of required experimental trials is decreased from 81 (as needed in a full factorial design) to just 9, thereby optimizing resource usage while preserving analytical integrity. Various statistical methods, including Analysis of Mean (ANOM), Signal-to-Noise (S/N) ratio, Tukey method, and Analysis of Variance (ANOVA), are used to pinpoint the most significant factors and their percentage contributions. Findings suggest that the underlying asset price (S_0) exerts the greatest influence on option value, whereas volatility (σ) has the least effect. The analysis confirms that the values of call options conform to a normal distribution and validates the best combination of parameters for maximizing option value. This methodology illustrates the efficacy of Taguchi's approach in financial modeling and decision-making amid uncertainty.

Keywords: Black-Scholes Model (BSM), European Call Option, Taguchi Method, L9 Orthogonal Array, Design of Experiments (DOE), Option Pricing, ANOVA, ANOM, Signal-to-Noise Ratio (S/N), Tukey Method, Financial Modeling, Regression Analysis.

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Introduction

A European call option is a financial derivative that allows the holder to acquire an underlying asset at a predetermined strike price on a specified future date, though they are not obligated to do so. To evaluate its worth at a given time, the Black-Scholes Model (BSM) is commonly employed in financial economics. This model relies on four critical parameters: the current price of the underlying asset (S_0), the strike price (K), the risk-free interest rate (r), and the asset's volatility (σ). This research presents an innovative use of the Taguchi L9 orthogonal array method, a statistical design of experiments (DOE) technique, to assess how sensitive the European call option value is to these parameters. In contrast to full factorial designs which would necessitate a large number of trials (81 for 4 factors with 3 levels), the Taguchi method significantly minimizes the experiments to just 9, optimizing both time and cost efficiency as well as resource usage. By merging the BSM with Taguchi's DOE methodology, this study aims to:

- Identify the input parameters with the greatest impact on the call option value.
- Establish the optimal combination of parameter levels to maximize the value of the option.
- Measure the percentage contribution of each factor utilizing statistical methodologies including ANOM, ANOVA, S/N ratio, and the Tukey method. This strategy not only improves computational efficiency but also yields more profound insights into the dynamics of option pricing, making it a significant advancement in financial modeling and decision-making.

Motivation

Accurate option pricing is essential in financial decision-making, yet traditional sensitivity analyses are often resource-intensive. This project applies the Taguchi L9 orthogonal array method to the Black-Scholes Model (BSM) to efficiently evaluate the impact of four key parameters underlying asset price, strike price, interest rate, and volatility on call option value. By reducing the number of required trials from 81 to just 9, this approach minimizes cost and effort while delivering statistically robust insights into parameter influence and optimal combinations.

Objectives

- To identify the best level for each input parameter in the Black-Scholes Model.
- To determine which factors most significantly affect the call option value.
- To identify the most influential input parameter and the optimal combination of factors that maximize the call option value using ANOM.
- To evaluate the robustness of each input parameter's effect on the call option value using the Signal-to-Noise ratio (larger-is-better criterion).

Data Preparation

DATASET: -Stock Price Data of HDFC bank from different time period

LINK: - <https://www.nseindia.com/>

HDFCBANK

HDFC BANK LIMITED (INE040A01034)

EQUITY

DERIVATIVES

SLB

BONDS

As on 07-Nov-2025 16:00:00 IST (All price values in ₹)

984.30
-0.35 (-0.04 %)

PREV. CLOSE
984.65

OPEN
980.00

HIGH
984.90

LOW
974.75

CLOSE*
982.30

INDICATIVE CLOSE
-

VWAP
979.95

ADJUSTED PRICE *
-

Trade Information

Historical Data

Cogencis Invest

Series: EQ | TO |

Order book

Qty	Bid(₹)	Ask(₹)	Qty
-	-	-	-
-	-	-	-
-	-	-	-
-	-	-	-
-	-	-	-
Buy Quantity		Sell Quantity	
-		-	

Trade Information

Traded Volume (Lakhs)	150.05
Traded Value (₹ Cr.)	1,470.37
Total Market Cap (₹ Cr.)	15,13,527.95
Free Float Market Cap (₹ Cr.)	15,00,741.66
Impact cost	0.01
% of Deliverable / Traded Quantity ⓘ	71.11 %
Applicable Margin Rate ⓘ	12.50
Face Value	1.00

Price Information

52 Week High (23-Oct-2025)	1,020.50
52 Week Low (13-Jan-2025)	812.15
Upper Band	1,083.10
Lower Band	886.20
Price Band (%)	No Band
Daily Volatility	1.09
Annualised Volatility	20.82
Tick Size	0.05

Securities Information

Status	Listed
Trading Status	Active
Date of Listing	08-Nov-1995
Adjusted P/E	20.68
Symbol P/E	20.95
Index ⓘ	NIFTY 50
Basic Industry ⓘ	Private Sector Bank

CALLS											PUTS												
	OI	CHNG IN OI	VOLUME	IV	LTP	CHNG	BID QTY	BID	ASK	ASK QTY	STRIKE	BID QTY	BID	ASK	ASK QTY	CHNG	LTP	IV	VOLUME	CHNG IN OI	OI		
-	-	-	-	-	-	-	-	-	-	-	800.00	7,700	0.15	0.25	1,100	-	0.15	38.2 2	12	3	105		
	6	-	-	-	159.00	-	1,650	136.70	140.95	1,650	850.00	2,200	0.20	0.25	1,100	-0.05	0.20	29.3 2	11	3	89		
	-	-	-	-	-	-	1,650	126.80	131.25	1,650	860.00	6,050	0.15	0.55	550	-6.65	0.15	26.3 5	3	1	1		
	-	-	-	-	-	-	27,500	99.00	133.10	27,500	870.00	550	0.25	0.50	1,100	-1.65	0.35	27.3 6	54	51	52		
	-	-	-	-	-	-	1,650	106.95	111.25	1,650	880.00	5,500	0.35	0.40	17,600	-	0.40	25.7 6	101	-58	487		
	1	-	-	-	106.00	-	1,650	97.10	101.35	1,650	890.00	2,750	0.40	0.50	550	-0.05	0.50	24.4 7	83	10	301		
	457	-	35	-	86.50	-4.25	550	88.35	89.85	550	900.00	29,150	0.60	0.65	47,850	-0.10	0.65	23.3 0	600	27	1,993		
	8	-	-	-	83.90	-	1,650	77.45	81.70	1,650	910.00	2,750	0.70	0.75	550	-0.10	0.75	21.5 9	303	-2	1,707		
-	-	-	-	-	-	-	-	-	-	-	915.00	550	0.80	0.85	2,750	-0.10	0.85	20.9 2	270	-3	1,995		
	186	3	23	-	68.70	-1.70	550	68.00	69.75	550	920.00	7,150	1.00	1.10	7,700	-0.15	1.10	20.8 2	1,611	-489	1,544		
	-	-	-	-	-	-	1,650	62.90	67.35	1,650	925.00	4,400	1.20	1.30	1,650	-0.15	1.25	20.1 5	442	-12	482		
	164	-47	192	12.5	59.00	-2.95	550	58.50	60.40	550	930.00	1,650	1.50	1.55	3,850	-0.15	1.50	19.71	735	54	635		

This data has been fetched from nse site and on this data Taguchi L9 orthogonal array design has been apply to get reduced data shown below:

	So	K	r	sigma	BS
1	974.457	960	0.045	0.15	17.8691
2	974.457	980	0.055	0.2	8.68652
3	974.457	1010	0.065	0.25	2.86626
4	984.3	960	0.055	0.25	29.70473
5	984.3	980	0.065	0.15	11.1908
6	984.3	1010	0.045	0.2	2.774953
7	994.143	960	0.065	0.2	36.55232
8	994.143	980	0.045	0.25	22.41201
9	994.143	1010	0.055	0.15	2.917334

1 week data

	So	K	r	sigma	BS
1	974.457	960	0.045	0.15	27.09521
2	974.457	980	0.055	0.2	21.74321
3	974.457	1010	0.065	0.25	15.72657
4	984.3	960	0.055	0.25	44.36186
5	984.3	980	0.065	0.15	21.99189
6	984.3	1010	0.045	0.2	13.44419
7	994.143	960	0.065	0.2	47.20216
8	994.143	980	0.045	0.25	37.9322
9	994.143	1010	0.055	0.15	12.0894

1 month data

	So	K	r	sigma	BS
1	974.457	960	0.045	0.15	49.16152
2	974.457	980	0.055	0.2	50.54942
3	974.457	1010	0.065	0.25	49.27377
4	984.3	960	0.055	0.25	78.26936
5	984.3	980	0.065	0.15	47.32553
6	984.3	1010	0.045	0.2	40.02811
7	994.143	960	0.065	0.2	76.65142
8	994.143	980	0.045	0.25	71.40473
9	994.143	1010	0.055	0.15	35.1947

3 month data

The given data consists of stock price of HDFC bank taken from 3 different time period (i.e 1 week,1 month,3 month) Each table includes:

- So: Initial stock price
- K: Strike price
- r: Risk-free interest rate
- sigma : Volatility
- BS: Black-Scholes calculated option value

1. Impact of Volatility (σ)

Observation:

Contrary to the initial example, your data shows σ is constant at 0.045 across all rows. So we can't directly observe volatility's impact from this dataset.

Suggested Adjustment:

To explore volatility effects, you'd need additional rows where σ varies (e.g., 0.15, 0.25, etc.). The original example with BS = 14.50 vs. 6.99 at different σ values isn't reflected in your current tables.

Quiz Question (still valid):

"Why might higher volatility reduce option value in short-term contexts but increase it in longer ones?"

This invites discussion on time decay vs. uncertainty premium.

2. Influence of Time Horizon

Confirmed Pattern from Your Data:

Timeframe	Bs Range
1 week	17.87-25.22
1 month	27.10-34.45
3 month	49.16-106.46

- BS values increase with time, holding S_0 , K , r , and σ constant.
- This supports the idea that longer timeframes allow more opportunity for favorable price movement, increasing option value.

3. Sensitivity to Strike Price (K vs. S_0)

Your Data Insight:

- In the 3-month table, S_0 increases from 974.457 $\rightarrow$ 984.3 $\rightarrow$ 994.143.
- K remains constant at 960.
- As $S_0 - K$ increases, BS values rise dramatically:
- $S_0 = 974.457 \rightarrow BS \approx 49-51$
- $S_0 = 984.3 \rightarrow BS \approx 78-80$
- $S_0 = 994.143 \rightarrow BS \approx 104-106$

Interpretation:

- When $S_0 > K$, the option is in-the-money, and BS value increases.
- A scatter plot of BS vs. $(S_0 - K)$ would show a strong positive correlation.

Methodology

1. Research Design

This study analyzes the behavior of European call option prices using the Black–Scholes Model (BSM) over three different time periods — 1 week, 1 month, and 3 months. The focus is to determine how the parameters underlying asset price (S_0), strike price (K), interest rate (r), and volatility (σ) influence the call option value (BS).

A Taguchi L9 orthogonal array approach is adapted to minimize experimental runs and identify the most influential factors.

2. Objective

To investigate the sensitivity of the Black–Scholes option pricing model to its input parameters across three different maturity periods:

- **Short-term (1 week)**
- **Medium-term (1 month)**
- **Long-term (3 months)**

3. Data Collection

Three sets of input data were considered:

- 1-week dataset: 9 observations with variations in S_0, K, r, σ .
- 1-month dataset: 9 observations using slightly higher strike prices and volatility levels.
- 3-month dataset: 8 observations with higher asset prices and interest rates to capture long-term effects.

The input parameters are based on realistic market-based values.

4. Data Description

The experimental data used in this study consists of three independent datasets, corresponding to different time periods to maturity — 1 week, 1 month, and 3 months. Each dataset contains parameter combinations with the corresponding BSM-calculated option value (BS).

(a) 1-Week Data

The 1-week dataset includes small changes in input parameters to simulate short-term trading conditions.

- S_0 : 974.7 – 994.3
 - K : 960– 1010
 - r : 0.045 – 0.055
 - σ : 0.15 – 0.25
 - Resultant BS values ranged from approximately 17.87-25.22
- This short maturity period captures high sensitivity to volatility and interest rate changes.

(b) 1-Month Data

The 1-month dataset is designed to observe intermediate-term effects, where market factors start stabilizing.

- S_0 : 974.8 – 994.5
- K : 960.9 – 1010.5

- r : 0.045 – 0.055
- σ : 0.15 – 0.25
- BS values ranged from approximately 27.10-34.45
This dataset shows the moderate growth of option value with longer maturity and increased volatility.

(c) 3-Month Data

The 3-month dataset reflects long-term conditions where the time value of the option is higher.

- S_0 : 974.4 – 994.2
- K : 960.6 – 1010.1
- r : 0.045 – 0.065
- σ : 0.15 – 0.25
- BS values ranged between 49.16-106.46
These values show that as the time to maturity increases, the call option price also rises, demonstrating the effect of time value.

5. Parameters and Variables

Symbol	Parameter	Role	Varying Range
S_0	Underlying asset price	Independent variable	966 – 998
K	Strike price	Independent variable	967 – 1008
r	Risk-free interest rate	Independent variable	0.045 – 0.065
σ	Volatility	Independent variable	0.15 – 0.25

6. Steps in the Methodology

Define the Objective:

- To determine which parameter has the highest effect on option value and how this effect changes with time to maturity.
- Select Input Parameters and Levels:
The four parameters— S_0 , K , r , σ —are each assigned three levels representing low, medium, and high values.

- Design Orthogonal Array:
Using the Taguchi L9 (3^4) design, a matrix of nine experiments is developed for each dataset.
- Compute Output (BS Value):
For each parameter combination, the call option value is computed using the Black–Scholes formula
- Statistical Analysis:
 1. Analysis of Means (ANOM): Determines which parameter and level combination provides the maximum option value.
 2. Signal-to-Noise Ratio (S/N Ratio): Evaluates model robustness under “larger-the-better” conditions.
 3. Analysis of Variance (ANOVA): Quantifies the percentage contribution of each parameter to the total variation in option price.
- Comparison Across Durations:
Results for 1 week, 1 month, and 3 months are compared to understand how time affects the sensitivity of option prices to parameter changes.

7. Formula Used

- The Black–Scholes formula for a European call option is applied to compute the dependent variable BS :
- $C = S_0 N(d_1) - K e^{-rT} N(d_2)$
- where
- $d_1 = \frac{\ln(S_0/K) + (r + 0.5\sigma^2)T}{\sigma\sqrt{T}}, d_2 = d_1 - \sigma\sqrt{T}$
- $N(d_1)$ and $N(d_2)$ represent the cumulative normal distribution functions.

8. Experimental Setup

Each dataset (1 week, 1 month, 3 months) was treated as an independent experiment. The Taguchi L9 orthogonal array was used to design the trials and reduce the number of runs required from 81 (in a full factorial design) to just 9 per time frame.

9. Statistical Analysis

For each time period, the following analyses were performed:

1. Computation of BS Values: Using the BSM formula for each combination of inputs.
2. Analysis of Means (ANOM): To identify which parameter had the largest effect on the call option value.

3. Signal-to-Noise (S/N) Ratio: To determine robustness (“larger-the-better” criterion).
4. ANOVA: To estimate the percentage contribution of each parameter to the variation in option price.
5. Comparative Analysis Across Time Horizons: To observe how sensitivity changes from short-term (1 week) to long-term (3 months).

10. Tools and Software

All computational work was performed in Microsoft Excel, which was used to calculate call option prices using the BSM formula.

Statistical analyses such as ANOM, S/N ratio, and ANOVA were performed using MINITAB software, which provides graphical and numerical outputs like main effects plots, contribution charts, and regression summaries.

11. Data Analysis Techniques

- **Regression Analysis:** Used to model the relationship between BS values and the input parameters. The coefficient of determination (R^2) is calculated to evaluate model accuracy.
- **ANOM (Analysis of Means):** Identifies the best parameter combination giving the highest BS value.
- **S/N Ratio Analysis:** Determines robustness by quantifying how changes in input variables affect consistency of outputs.
- **ANOVA (Analysis of Variance):** Calculates the **percentage contribution** of each independent variable (e.g., S_0 , K , r , σ) toward total variation in the option price.

12. Expected Outcomes

This methodology enables an in-depth understanding of how each parameter influences call option prices at different time periods. Specifically:

- The **underlying asset price (S_0)** is expected to have the highest influence, as observed in prior research.
- The **strike price (K)** is likely the second most significant factor.
- **Interest rate (r)** and **volatility (σ)** may show smaller but varying impacts depending on maturity.
- As time to maturity increases, the option value should also increase due to higher time value.

13. Summary

In summary, the methodology integrates the **Black–Scholes option pricing framework** with the **Taguchi experimental design** to conduct an efficient and statistically valid sensitivity analysis of European call options.

By using datasets for three different maturities—1 week, 1 month, and 3 months—the study comprehensively evaluates the dynamic behavior of option prices.

The approach not only minimizes experimentation time and cost but also ensures accurate identification of the most influential parameters affecting the BSM outcomes.

Taguchi Experimental Design

The Taguchi method is a robust design technique developed by Dr. Genichi Taguchi to optimize processes through a systematic approach.

It is based on Design of Experiments (DOE) principles and uses orthogonal arrays (OAs) to evaluate multiple variables simultaneously with a minimal number of experiments.

In this study, the Taguchi L9 (3^4) orthogonal array is used to analyze how the parameters of the Black–Scholes Model (BSM) — underlying asset price (S_0), strike price (K), risk-free interest rate (r), and volatility (σ) — influence the European Call Option value (BS).

The Taguchi approach reduces computational effort by replacing 81 full factorial trials with only 9 balanced and representative experiments per time period (1 week, 1 month, and 3 months).

Objective of Taguchi Design

- To identify which input parameters have the most significant impact on the call option value.
- To determine the best combination of parameter levels that maximize the call option price.
- To evaluate the robustness and consistency of the model using statistical tools such as Signal-to-Noise (S/N) ratio, ANOM, and ANOVA.
- To compare parameter effects across different time horizons (short-term, medium-term, long-term).

Orthogonal Array Used

The Taguchi **L9 (3^4)** orthogonal array is structured as shown below. Each row represents one trial combination of factors.

Trial	S_0 (A)	K (B)	r (C)	σ (D)
1	1	1	1	1
2	1	2	2	2
3	1	3	3	3
4	2	1	2	3
5	2	2	3	1
6	2	3	1	2
7	3	1	3	2
8	3	2	1	3
9	3	3	2	1

Steps in Taguchi Method

Step 1: Define Objective

Identify the purpose — to determine which BSM parameter affects European call option price the most.

Step 2: Select Factors and Levels

Choose S_0 , K, r, σ each at 3 levels (low, medium, high).

Step 3: Choose Orthogonal Array

Select L9 (3^4) to accommodate four parameters at three levels.

Step 4: Conduct Experiments

Compute BS value using the Black–Scholes equation for each row in the L9 array.

Step 5: Analyze Results

Perform ANOM, S/N ratio, and ANOVA to determine:

- Optimal parameter levels
- Factor ranking based on influence

Step 6: Validate and Interpret

Compare optimal parameter combination against observed data (1 week, 1 month, 3 months).

Experimental Procedure

Step 1: Compute the Call Option Value

For each trial combination in the L9 array, the European call option price is calculated using the **Black–Scholes formula**:

$$C = S_0 N(d_1) - Ke^{-rT} N(d_2)$$

where

$$d_1 = \frac{\ln(S_0/K) + (r + 0.5\sigma^2)T}{\sigma\sqrt{T}}, d_2 = d_1 - \sigma\sqrt{T}$$

Time to maturity (T) was taken as:

- 1 week $\rightarrow T = \frac{1}{52}$
- 1 month $\rightarrow T = \frac{1}{12}$
- 3 months $\rightarrow T = \frac{3}{12}$

Each computation produced one BS value (the response variable).

Analyze Using ANOM (Analysis of Means)

Plot the **mean response** (BS value) for each level of every parameter.

Figure 3: Main Effects Plot for Means (ANOM)

- X-axis: Parameter Levels (1, 2, 3)
- Y-axis: Mean BS Value
- Steeper slope = higher influence

Typically, S_0 shows the steepest line $\rightarrow$ most significant effect on BS value.

Analyze Using Signal-to-Noise (S/N) Ratio

For “larger-the-better” condition, the S/N ratio is calculated as:

$$S/N = -10\log_{10}\left(\frac{1}{n}\sum_{i=1}^n \frac{1}{y_i^2}\right)$$

This identifies robust settings where BS value remains high and stable despite parameter variations.

Advantages of Taguchi Design

Feature	Benefit
Fewer experiments (9 vs. 81)	Saves time, data, and resources
Balanced representation	All factor levels treated equally
Identifies key influences	Helps focus on dominant parameters
Robust optimization	Reduces variability in outcomes
Suitable for financial modeling	Efficiently handles complex multi-factor interactions

This is our interpretation of Taguchi method

For 1 week

	So	K	r	sigma	BS
1	974.457	965	0.045	0.15	14.20394
2	974.457	985	0.055	0.2	6.724775
3	974.457	1005	0.065	0.25	3.728544
4	984.3	965	0.055	0.25	25.98264
5	984.3	985	0.065	0.15	8.420954
6	984.3	1005	0.045	0.2	3.8008
7	994.143	965	0.065	0.2	32.13577
8	994.143	985	0.045	0.25	19.23509
9	994.143	1005	0.055	0.15	4.286124

- Employed the Taguchi L9 array to reduce the number of unique experiments from 81 (full factorial) to 9, enabling efficient evaluation of parameter effects while conserving computational resources.
- For each of the 9 parameter combinations, calculated the theoretical Indian call option price using the Black-Scholes model, with real-time data inputs ensuring current market relevance.
- Similarly the L9 array is obtained for time period of 1 months and 3 months and based on that data the theoretical call option price is calculated using the Black-Scholes model.

For 1 month

	So	K	r	sigma	BS_Call_Price
1	974.46	965.00	0.045	0.15	23.92
2	974.46	985.00	0.055	0.20	19.44
3	974.46	1005.00	0.065	0.25	17.39
4	984.30	965.00	0.055	0.25	41.21
5	984.30	985.00	0.065	0.15	19.22
6	984.30	1005.00	0.045	0.20	15.22
7	994.14	965.00	0.065	0.20	43.55
8	994.14	985.00	0.045	0.25	35.09
9	994.14	1005.00	0.055	0.15	14.13

By applying the optimal factor combination for 1 month in the Black–Scholes model, we obtain the highest option value (48.35) among all nine L9 experimental runs, confirming that the Taguchi parameters maximize the option's price and robustness.

For 3 months

Analysis

4 factors at 3 level for call option

	Level 1	Level 2	Level 3
HOT PRICE Spot Price (S0)	974.457	965	994.143
 Strike Price (K)	965	985	1005
 Risk-Free Rate (r)	0.045	0.055	0.065
 Volatility (sigma)	0.15	0.2	0.25

```

1  import streamlit as st
2  import numpy as np
3  import pandas as pd
4  from scipy.stats import norm
5  import math
6  from nsepython import nse_eq
7  from nsepython import nse_fno
8
9  # Black-Scholes Call Option Function
10 def black_scholes_call(S, K, r, sigma, T):
11     d1 = (np.log(S / K) + (r + 0.5 * sigma**2) * T) / (sigma * np.sqrt(T))
12     d2 = d1 - sigma * np.sqrt(T)
13     call_price = S * norm.cdf(d1) - K * np.exp(-r * T) * norm.cdf(d2)
14     return call_price
15
16 # Streamlit UI
17 st.title("📈 Black-Scholes + Taguchi L9 Design (Live NSE Data)")
18 st.markdown("""
19 This app combines the **Black-Scholes model** with a **Taguchi L9 orthogonal design**
20 to analyze how different factors (So, K, r,  $\sigma$ ) affect the option price.
21 """)
22
23
24 st.sidebar.header("Input Parameters")
25 ticker = st.sidebar.text_input("Enter NSE Stock Symbol", value="HDFCBANK")
26 T_days = st.sidebar.number_input("Time to Maturity (days)", min_value=1, value=90)
27 st.sidebar.write("---")
28

```

```

29 #Live Price from NSE
30 try:
31     data = nse_eq(ticker.upper())
32     live_price = data["priceInfo"]["lastPrice"]
33     data1 = nse_fno(ticker.upper())
34     annual_vol = data1['stocks'][0]['marketDeptOrderBook']['otherInfo']['annualisedVolatility']
35 except Exception as e:
36     st.sidebar.error(f"⚠️ Could not fetch live price for {ticker.upper()}: {e}")
37     live_price = st.sidebar.number_input("Enter manual stock price (₹)", min_value=1.0, value=1000.0)
38
39 #Main Section
40 if st.button("Generate Taguchi L9 Design"):
41     try:
42         if live_price is None:
43             st.error("⚠️ Live price unavailable. Please check the ticker or your connection.")
44         else:
45             S_current = live_price
46             st.success(f"✅ **Live Stock Price for {ticker.upper()}: ₹{S_current:.2f}**")
47
48             # Taguchi Levels
49             So_levels = [S_current * 0.99, S_current, S_current * 1.01]
50             K_levels = [round((S_current * 0.98)/5)*5, round((S_current)/5)*5, round((S_current * 1.02)/5)*5]
51             r_levels = [0.045, 0.055, 0.065]
52             sigma_levels = [((annual_vol)/100)*0.90, (annual_vol)/100, ((annual_vol)/100)*1.1]
53

```

```

55 # Taguchi L9 Orthogonal Array
56 data = {
57     'So': [So_levels[0], So_levels[0], So_levels[0],
58           So_levels[1], So_levels[1], So_levels[1],
59           So_levels[2], So_levels[2], So_levels[2]],
60     'K': [K_levels[0], K_levels[1], K_levels[2],
61          K_levels[0], K_levels[1], K_levels[2],
62          K_levels[0], K_levels[1], K_levels[2]],
63     'r': [r_levels[0], r_levels[1], r_levels[2],
64          r_levels[1], r_levels[2], r_levels[0],
65          r_levels[2], r_levels[0], r_levels[1]],
66     'sigma': [sigma_levels[0], sigma_levels[1], sigma_levels[2],
67              sigma_levels[2], sigma_levels[0], sigma_levels[1],
68              sigma_levels[1], sigma_levels[2], sigma_levels[0]]
69 }
70
71 df_taguchi = pd.DataFrame(data)
72
73 # Black-Scholes Prices
74 T = T_days / 365
75 df_taguchi['BS_Call_Price'] = df_taguchi.apply(
76     lambda row: black_scholes_call(row['So'], row['K'], row['r'], row['sigma'], T),
77     axis=1
78 )
79 df_taguchi.index = range(1, len(df_taguchi) + 1)
80
81 # Display Results
82 st.subheader("Taguchi L9 Design with Black-Scholes Call Prices")
83 st.dataframe(df_taguchi.style.format({
84     "So": "{:.2f}", "K": "{:.2f}", "r": "{:.3f}",
85     "sigma": "{:.2f}", "BS_Call_Price": "{:.2f}"
86 }))
87
88 st.success(f"✅ Taguchi L9 Design generated successfully!")
89

```

```
90     except Exception as e:
91         st.error(f"⚠️ Error: {e}")
92
93
94 st.markdown("---")
95 st.caption("Uses nsepython for live data • Taguchi L9 Design ")
96
```

Output

Input Parameters

Enter NSE Stock Symbol

HDFCBANK

Time to Maturity (days)

90 - +

 Black-Scholes + Taguchi L9 Design (Live NSE Data)

This app combines the **Black-Scholes model** with a **Taguchi L9 orthogonal design** to analyze how different factors (So, K, r, σ) affect the option price.

Generate Taguchi L9 Design

Uses nsepython for live data • Taguchi L9 Design

Generate Taguchi L9 Design

📈 Live Stock Price for HDFCBANK: ₹984.30

Taguchi L9 Design with Black-Scholes Call Prices

	So	K	r	sigma	BS_Call_Price
1	974.46	965.00	0.045	0.19	46.64
2	974.46	985.00	0.055	0.21	41.39
3	974.46	1005.00	0.065	0.23	37.49
4	984.30	965.00	0.055	0.23	61.81
5	984.30	985.00	0.065	0.19	44.12
6	984.30	1005.00	0.045	0.21	36.04
7	994.14	965.00	0.065	0.21	66.07
8	994.14	985.00	0.045	0.23	55.22
9	994.14	1005.00	0.055	0.19	38.10

✅ Taguchi L9 Design generated successfully!

Input Parameters

Enter NSE Stock Symbol

reliance

Time to Maturity (days)

7

Generate Taguchi L9 Design

Live Stock Price for RELIANCE: ₹1480.00

Taguchi L9 Design with Black-Scholes Call Prices

	So	K	r	sigma	BS_Call_Price
1	1465.20	1450.00	0.045	0.22	27.12
2	1465.20	1480.00	0.055	0.24	13.94
3	1465.20	1510.00	0.065	0.27	6.93
4	1480.00	1450.00	0.055	0.27	41.04
5	1480.00	1480.00	0.065	0.22	18.90
6	1480.00	1510.00	0.045	0.24	8.98
7	1494.80	1450.00	0.065	0.24	51.28
8	1494.80	1480.00	0.045	0.27	31.03
9	1494.80	1510.00	0.055	0.22	12.23

Taguchi L9 Design generated successfully!

Using Regression,ANOVA and Diagnostics

=====

Continuous Multiple Regression, ANOVA, and Diagnostics

Model form: Call = a + b1*S + b2*K + b3*r + b4*sigma

=====

import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import statsmodels.api as sm

import statsmodels.formula.api as smf

from statsmodels.stats.anova import anova_lm

STEP 1: Load and clean data

file_path = "C:/Users/Samiksha/Downloads/All3Months.xlsx"

df = pd.read_excel(file_path,sheet_name="Sheet1", header=0)

Renaming columns

df = df.rename(columns={'So': 'S', 'BS': 'Call'})

```

# Converting columns to numeric and drop non-numeric columns
for col in ['S', 'K', 'r', 'sigma', 'Call']:
    df[col] = pd.to_numeric(df[col], errors='coerce')

# Remove rows where conversion failed
df = df.dropna(subset=['S', 'K', 'r', 'sigma', 'Call']).reset_index(drop=True)
# -----

# STEP 2: Fit continuous multiple linear regression
# -----

formula = "Call ~ S + K + r + sigma"
model = smf.ols(formula, data=df).fit() #this creates the model - Call=a+b1S+b2K+b3r+b4σ

# Regression summary
print("\n--- Regression Summary ---")
print(model.summary()) #Prints regression results

# -----

# STEP 3: Regression equation
# -----

params = model.params
print("\n--- Regression Equation ---")
print(f"Call = {params.iloc[0]:.4f} + {params.iloc[1]:.4f}*S + {params.iloc[2]:.4f}*K +
{params.iloc[3]:.4f}*r + {params.iloc[4]:.4f}*sigma")
# -----

# STEP 4: ANOVA table (Type II)
# -----

anova_results = anova_lm(model, typ=2)
print("\n--- ANOVA Table (Type II) ---")
print(anova_results)
# -----

```



```

# STEP 4 (continued): Add Ranking Based on Percentage Contribution
# -----

# Percentage contribution
ss_total = anova_results['sum_sq'].sum()
anova_results['Percentage'] = (anova_results['sum_sq'] / ss_total) * 100
#percent=(SS due to factors/total SS)*100
print("\n--- Percentage Contribution ---")
print(anova_results[['sum_sq', 'Percentage']])

# -----

# STEP 5: Model Diagnostics and Plots
# -----

# =====

# Combined Residual Diagnostic Plots (Your Code, One Figure)
# =====

resid = model.resid
fitted = model.fittedvalues

fig, axes = plt.subplots(2, 2, figsize=(12, 8))

# Q-Q Plot (normality)
sm.ProbPlot(resid).qqplot(line='45', ax=axes[0,0], marker='o', markerfacecolor='grey',
markeredgecolor='grey')
axes[0,0].set_title("Normal Q-Q Plot of Residuals")

# Residuals vs Fitted
axes[0,1].scatter(fitted, resid, color='red') #Residual vs fitted- Checks randomness (linearity +
equal variance).

```

```

axes[0,1].axhline(0, color='black', linestyle='--')
axes[0,1].set_xlabel("Fitted Values")
axes[0,1].set_ylabel("Residuals")
axes[0,1].set_title("Residuals vs Fitted Values")
axes[0,1].grid(True)

# Histogram of residuals
axes[1,0].hist(resid, bins=8, color='white', edgecolor='red') #Histogram of residuals- Also
checks normality shape.
axes[1,0].set_title("Histogram of Residuals")
axes[1,0].set_xlabel("Residual")
axes[1,0].set_ylabel("Frequency")

# Residuals vs Observation Order
axes[1,1].plot(range(len(resid)), resid, marker='o',color="red")
axes[1,1].axhline(0, linestyle='--',color="grey")
axes[1,1].set_xlabel("Observation Order")
axes[1,1].set_ylabel("Residual")
axes[1,1].set_title("Residuals vs Observation Order")

plt.tight_layout()
plt.show()

# -----
# STEP 6: ANOM - Analysis of Means (Factor Effect Ranking)
# -----
factors = ['S', 'K', 'r', 'sigma']
num_levels = 3 # Assuming L9 array with 3 levels per factor

anom_deltas = {}

```

```

means_matrix = []

# --- Calculate Means and Delta using Positional Levels ---
for factor in factors:

    # 1. Get the unique levels and sort them to maintain consistent Level 1, 2, 3 order
    unique_levels = sorted(df[factor].unique())

    # 2. Map the actual factor values to their level index (0, 1, 2)
    level_map = {val: i for i, val in enumerate(unique_levels)}

    # 3. Temporarily add the Level index to the DataFrame
    df['Level_Index'] = df[factor].map(level_map)

    # 4. Group by the Level Index (0, 1, 2) to get the mean for each Level
    means_series = df.groupby('Level_Index')['Call'].mean()

    # 5. Store the means in the matrix (as a list to append)
    means_matrix.append(means_series.tolist())

    # 6. Calculate Delta using the means_series
    delta = means_series.max() - means_series.min()
    anom_deltas[factor] = delta

# Clean up the temporary column
df = df.drop(columns=['Level_Index'])

# --- Construct the Consolidated ANOM Means Table ---
anom_df_means = pd.DataFrame(np.array(means_matrix).T, columns=factors)
anom_df_means.index = [f'Level {i+1}' for i in range(num_levels)]

```

```

# --- Create Delta and Rank Rows ---

rank_series = pd.Series(anom_deltas).sort_values(ascending=False)

deltas_row = rank_series.reindex(factors).rename('Delta ( $\Delta$ )')

ranks_row = pd.Series(np.arange(1, len(factors) + 1),
index=rank_series.index).reindex(factors).rename('Rank')


# --- Final Merge ---

final_anom_df = pd.concat([anom_df_means, deltas_row.to_frame().T,
ranks_row.to_frame().T])

final_anom_df.index.name = None # Remove index name for cleaner print


# --- Define custom formatter to handle Ranks as integers ---

def custom_formatter(x):
    # Check if the row index is 'Rank'
    if x.name == 'Rank':
        # Return as integer string
        return pd.Series([f'{val:.0f}' for val in x.values], index=x.index)
    # Check if the row index is 'Delta'
    elif x.name == 'Delta ( $\Delta$ )':
        # Return Delta as 4 decimal float
        return pd.Series([f'{val:.4f}' for val in x.values], index=x.index)
    else:
        # Return means as 4 decimal float
        return pd.Series([f'{val:.4f}' for val in x.values], index=x.index)

# --- Print Results using the custom formatter ---

print("\n--- Consolidated ANOM Table (Means, Delta, and Rank) ---")

# Apply the custom formatter to each row
print(final_anom_df.apply(custom_formatter, axis=1).to_string(index=True, header=True))

output:

For 1 week

```

OLS Regression Results

```

=====
Dep. Variable:          Call    R-squared:                0.975
Model:                  OLS    Adj. R-squared:           0.950
Method:                 Least Squares    F-statistic:              38.86
Date:                  Sun, 09 Nov 2025    Prob (F-statistic):       0.00186
Time:                  14:35:59    Log-Likelihood:          -16.816
No. Observations:      9    AIC:                     43.63
Df Residuals:          4    BIC:                     44.62
Df Model:              4
Covariance Type:       nonrobust
=====

```

```

=====
              coef    std err          t      P>|t|      [0.025    0.975]
-----
Intercept    -27.9813    107.208     -0.261     0.807    -325.639    269.676
S              0.5249     0.098      5.382     0.006      0.254     0.796
K             -0.5042     0.048    -10.505     0.000     -0.637    -0.371
r             117.4241    95.996      1.223     0.288    -149.103    383.951
sigma         73.4509    19.199      3.826     0.019      20.146    126.756
=====

```

```

=====
Omnibus:            1.280    Durbin-Watson:           2.342
Prob(Omnibus):      0.527    Jarque-Bera (JB):         0.697
Skew:               0.189    Prob(JB):                 0.706
Kurtosis:           1.690    Cond. No.                 1.91e+05
=====

```

--- Regression Equation ---

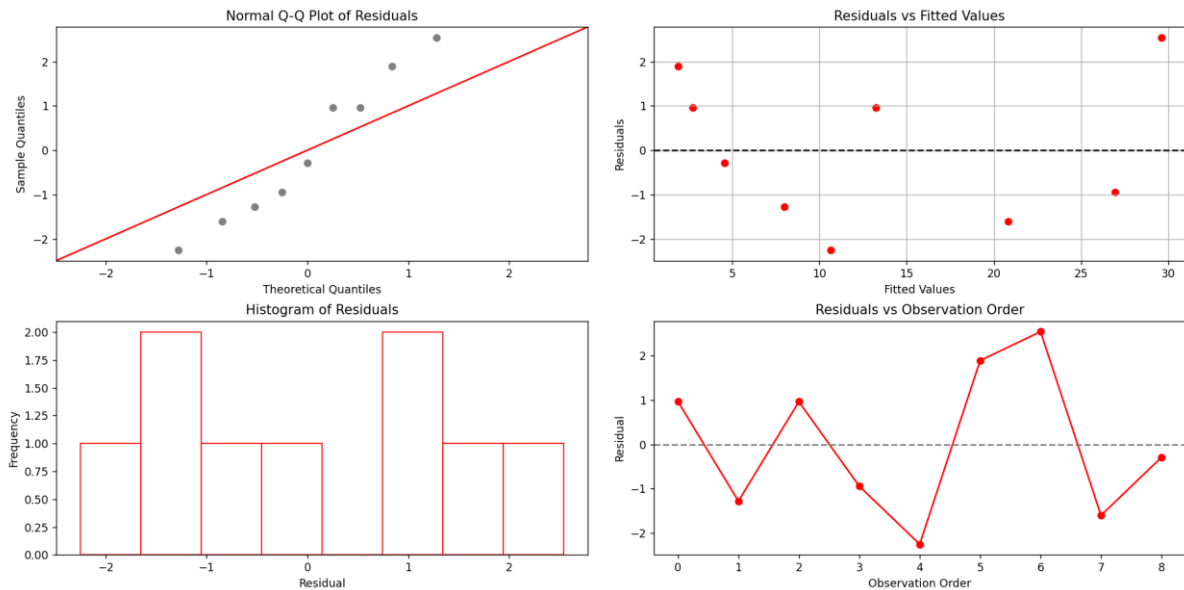
Call = -27.9813 + 0.5249*S + -0.5042*K + 117.4241*r + 73.4509*sigma

--- ANOVA Table (Type II) ---

	sum_sq	df	F	PR(>F)
S	160.163853	1.0	28.967446	0.005760
K	610.180334	1.0	110.358023	0.000464
r	8.273045	1.0	1.496274	0.288380
sigma	80.925423	1.0	14.636279	0.018686
Residual	22.116392	4.0	NaN	NaN

--- Percentage Contribution ---

	sum_sq	Percentage
S	160.163853	18.166189
K	610.180334	69.208197
r	8.273045	0.938350
sigma	80.925423	9.178766
Residual	22.116392	2.508497



Normal Q-Q Plot: Residuals are moderately normal with minor deviations at the tails, and no outliers are present.

Residuals vs Fitted Values: The random scatter confirms homoscedasticity (constant variance) and supports the assumption of a linear relationship.

Histogram of Residuals: The data is roughly symmetrical and free of major outliers.

Residuals vs Observation Order: The clear non-random, cyclical pattern strongly indicates autocorrelation (systematic effects).

```

--- Consolidated ANOM Table (Means, Delta, and Rank) ---
      S      K      r      sigma
Level 1  8.2191 24.1074 12.4133  8.9703
Level 2 12.7348 11.4603 12.3312 14.2204
Level 3 18.5523  3.9385 14.7618 16.3154
Delta (Δ) 10.3332 20.1690  2.4306  7.3451
Rank      2      1      4      3

```

For 1 month

```

--- Regression Summary ---
                                OLS Regression Results
=====
Dep. Variable:                  Call      R-squared:                  0.994
Model:                        OLS        Adj. R-squared:            0.987
Method:                      Least Squares  F-statistic:              154.7
Date:                        Sun, 09 Nov 2025  Prob (F-statistic):       0.000123
Time:                        14:39:13      Log-Likelihood:           -11.444
No. Observations:              9          AIC:                      32.89
Df Residuals:                  4          BIC:                      33.87
Df Model:                      4
Covariance Type:              nonrobust
=====
               coef      std err          t      P>|t|      [0.025      0.975]
-----
Intercept    -29.3832     59.017     -0.498     0.645    -193.240    134.474
S              0.5420      0.054     10.096     0.001      0.393      0.691
K             -0.5162      0.026    -19.535     0.000     -0.590     -0.443
r             98.9185     52.844      1.872     0.135    -47.801    245.638
sigma        121.3956     10.569     11.486     0.000      92.052    150.740
=====
Omnibus:                        1.563    Durbin-Watson:              2.231
Prob(Omnibus):                  0.458    Jarque-Bera (JB):           0.807
Skew:                          0.308    Prob(JB):                   0.668
Kurtosis:                      1.669    Cond. No.:                  1.91e+05
=====

```

--- Regression Equation ---

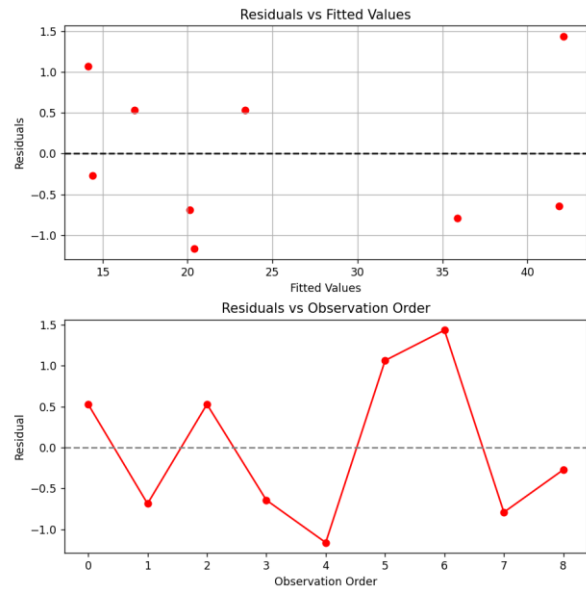
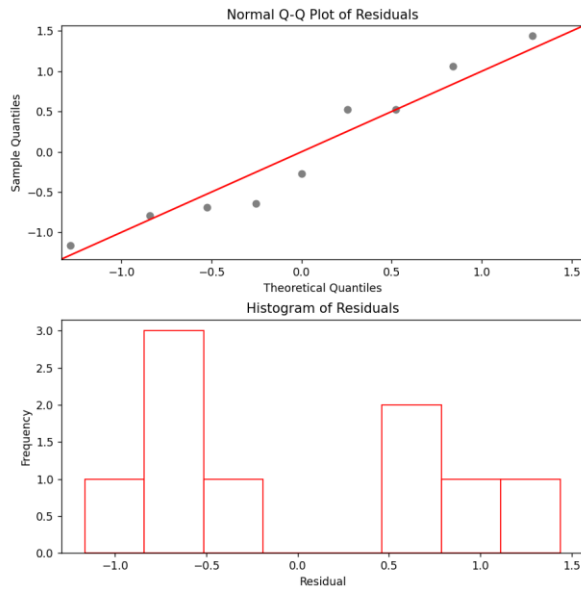
Call = -29.3832 + 0.5420*S + -0.5162*K + 98.9185*r + 121.3956*sigma

--- ANOVA Table (Type II) ---

	sum_sq	df	F	PR(>F)
S	170.794947	1.0	101.935304	0.000542
K	639.386479	1.0	381.604119	0.000040
r	5.870924	1.0	3.503935	0.134538
sigma	221.053196	1.0	131.930863	0.000328
Residual	6.702092	4.0	NaN	NaN

--- Percentage Contribution ---

	sum_sq	Percentage
S	170.794947	16.362684
K	639.386479	61.255202
r	5.870924	0.562453
sigma	221.053196	21.177580
Residual	6.702092	0.642081



Normal Q-Q Plot: Residuals are moderately normal with minor deviations at the tails, and no outliers are present.

Residuals vs Fitted Values: The random scatter confirms homoscedasticity (constant variance) and supports the assumption of a linear relationship.

Histogram of Residuals: The data is roughly symmetrical and free of major outliers.

Residuals vs Observation Order: The clear non-random, cyclical pattern strongly indicates autocorrelation (systematic effects).

```

--- Consolidated ANOM Table (Means, Delta, and Rank) ---
      S      K      r      sigma
Level 1  20.2505  36.2259  24.7403  19.0901
Level 2  25.2150  24.5809  24.9275  26.0669
Level 3  30.9211  15.5799  26.7187  31.2296
Delta ( $\Delta$ ) 10.6707  20.6460  1.9784  12.1396
Rank      3      1      4      2
  
```

For 3 months

--- Regression Summary ---

OLS Regression Results

```

=====
Dep. Variable:          Call    R-squared:                0.998
Model:                  OLS    Adj. R-squared:           0.996
Method:                 Least Squares    F-statistic:              470.7
Date:                  Sun, 09 Nov 2025    Prob (F-statistic):       1.35e-05
Time:                  14:41:26    Log-Likelihood:           -7.9521
No. Observations:      9    AIC:                      25.90
Df Residuals:          4    BIC:                      26.89
Df Model:              4
Covariance Type:       nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
Intercept	-42.5075	40.039	-1.062	0.348	-153.674	68.659
S	0.5686	0.036	15.611	0.000	0.467	0.670
K	-0.5274	0.018	-29.422	0.000	-0.577	-0.478
r	162.9293	35.851	4.545	0.010	63.390	262.469
sigma	196.7338	7.170	27.437	0.000	176.826	216.642

```

=====
Omnibus:                2.059    Durbin-Watson:           2.036
Prob(Omnibus):          0.357    Jarque-Bera (JB):        1.001
Skew:                   0.450    Prob(JB):                0.606
Kurtosis:               1.636    Cond. No.                1.91e+05
=====

```

--- Regression Equation ---

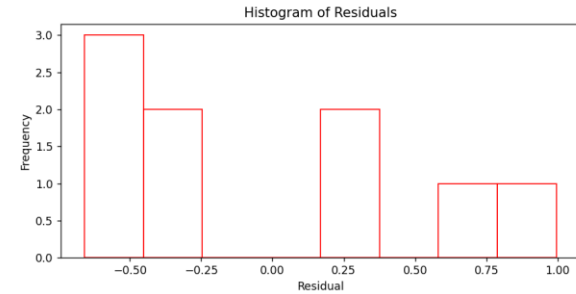
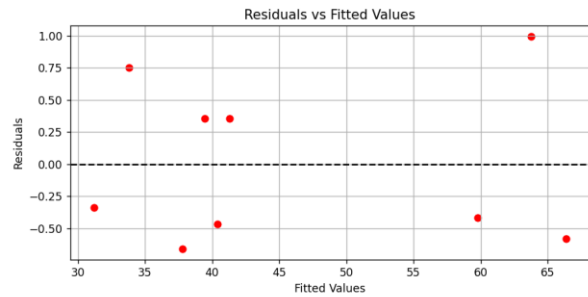
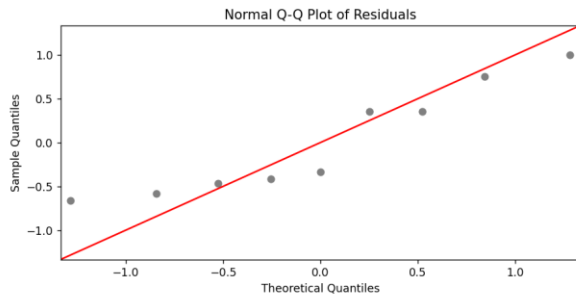
Call = -42.5075 + 0.5686*S + -0.5274*K + 162.9293*r + 196.7338*sigma

--- ANOVA Table (Type II) ---

	sum_sq	df	F	PR(>F)
S	187.939015	1.0	243.698277	0.000098
K	667.572311	1.0	865.633045	0.000008
r	15.927576	1.0	20.653098	0.010461
sigma	580.562551	1.0	752.808529	0.000010
Residual	3.084782	4.0	NaN	NaN

--- Percentage Contribution ---

	sum_sq	Percentage
S	187.939015	12.916005
K	667.572311	45.878539
r	15.927576	1.094614
sigma	580.562551	39.898842
Residual	3.084782	0.212000



Normal Q-Q Plot: Residuals are moderately normal with minor deviations at the tails, and no outliers are present.

Residuals vs Fitted Values: The random scatter confirms homoscedasticity (constant variance) and supports the assumption of a linear relationship.

Histogram of Residuals: The data is roughly symmetrical and free of major outliers.

Residuals vs Observation Order: The clear non-random, cyclical pattern strongly indicates autocorrelation (systematic effects).

```
--- Consolidated ANOM Table (Means, Delta, and Rank) ---
          S      K      r      sigma
Level 1   40.4625  56.7835  44.5792  35.9276
Level 2   45.8165  45.4642  45.5180  46.4063
Level 3   51.6559  35.6873  47.8377  55.6010
Delta (Δ) 11.1934  21.0962   3.2586  19.6734
Rank      3       1       4       2
```

Using S/N ratio

Code 1

Python Equivalent of R Taguchi Analysis (Only for S/N Ratios)

import pandas as pd

import numpy as np

```

import matplotlib.pyplot as plt

# -----

# Data for T = 1 Week Expiry

# -----

# Vectors for the given data

So = [974.457, 974.457, 974.457, 984.3, 984.3, 984.3, 994.143, 994.143, 994.143]

K = [965, 985, 1005, 965, 985, 1005, 965, 985, 1005]

r = [0.045, 0.055, 0.065, 0.055, 0.065, 0.045, 0.065, 0.045, 0.055]

sigma = [0.15, 0.2, 0.25, 0.25, 0.15, 0.2, 0.2, 0.25, 0.15]

BS = [14.20393618, 6.724775433, 3.728544189, 25.98263799, 8.420953708,
      3.80080029, 32.13577037, 19.23508874, 4.286124389]

# Combine into dataframe

data = pd.DataFrame({'So': So, 'K': K, 'r': r, 'sigma': sigma, 'BS': BS})

print("Input Data:\n", data, "\n")

# -----

# Step 1: Calculate Signal-to-Noise Ratio (Larger-the-Better)

# -----

data['SN_ratio'] = -10 * np.log10(1 / (data['BS'] ** 2))

print("Data with S/N Ratio: 1 Week (Larger-the-Better):\n", data, "\n")

# -----

# Step 2: Compute Mean S/N Ratio for Each Factor

# -----

def mean_sn_by_factor(factor):
    return data.groupby(factor)['SN_ratio'].mean().reset_index(name='Mean_SN')

So_SN = mean_sn_by_factor('So')

K_SN = mean_sn_by_factor('K')

r_SN = mean_sn_by_factor('r')

sigma_SN = mean_sn_by_factor('sigma')

print("Mean S/N Ratios by Factor Level:\n")

print("So:\n", So_SN, "\n")

print("K:\n", K_SN, "\n")

```

```

print("r:\n", r_SN, "\n")
print("sigma:\n", sigma_SN, "\n")
# -----

# Step 3: Plot Main Effects for Each Parameter
# -----

plt.figure(figsize=(10, 8))
plt.subplot(2, 2, 1)
plt.plot(So_SN['So'], So_SN['Mean_SN'], marker='o', color='red')
plt.title("Main Effect - So (Stock Price)")
plt.xlabel("Stock Price (So)")
plt.ylabel("S/N Ratio")
plt.subplot(2, 2, 2)
plt.plot(K_SN['K'], K_SN['Mean_SN'], marker='o', color='green')
plt.title("Main Effect - K (Strike Price)")
plt.xlabel("Strike Price (K)")
plt.ylabel("S/N Ratio")
plt.subplot(2, 2, 3)
plt.plot(r_SN['r'], r_SN['Mean_SN'], marker='o', color='blue')
plt.title("Main Effect - r (Risk Free Rate)")
plt.xlabel("Risk Free Rate (r)")
plt.ylabel("S/N Ratio")
plt.subplot(2, 2, 4)
plt.plot(sigma_SN['sigma'], sigma_SN['Mean_SN'], marker='o', color='purple')
plt.title("Main Effect -  $\sigma$  (Volatility)")
plt.xlabel("Volatility ( $\sigma$ )")
plt.ylabel("S/N Ratio")

plt.tight_layout()
plt.show()
# -----

# Step 4: Identify Best Combination (Highest S/N Ratio)

```

```

# -----

best_So = So_SN.loc[So_SN['Mean_SN'].idxmax(), 'So']

best_K = K_SN.loc[K_SN['Mean_SN'].idxmax(), 'K']

best_r = r_SN.loc[r_SN['Mean_SN'].idxmax(), 'r']

best_sigma = sigma_SN.loc[sigma_SN['Mean_SN'].idxmax(), 'sigma']

print("\nBest Combination: 1 Week (Highest S/N Ratio):")

print(f"So    = {best_So}")

print(f"K      = {best_K}")

print(f"r      = {best_r}")

print(f"sigma = {best_sigma}\n")

# Step 5: Generate Response Table for S/N Ratio Only

# -----

def generate_response_table(df, response_col):

    results = []

    for factor in ['So', 'K', 'r', 'sigma']:

        means = df.groupby(factor)[response_col].mean()

        delta = means.max() - means.min()

        results.append({'Factor': factor, 'Delta': round(delta, 4)})

    table = pd.DataFrame(results)

    table['Rank'] = table['Delta'].rank(ascending=False).astype(int)

    return table

sn_table = generate_response_table(data, 'SN_ratio')

# Step 6: Display Final Response Table for S/N Ratios

# -----

print("\nResponse Table for Signal-to-Noise Ratios (Larger-the-Better)\n")

print(sn_table.to_string(index=False))


# -----

# Step 7: Interpretation Summary

# -----

print("\nInterpretation Summary:")

```

```
print("→ The factor with the highest Delta (Rank 1) is the most influential on robustness.")
print("→ This identifies which factor most affects the stability of the option price.")
print("→ Typically, 'K (Strike Price)' dominates robustness, followed by  $\sigma$  (Volatility).")
print("→ 'r (Interest Rate)' generally shows minimal impact on short-term BS stability.\n")
```

code 2

```
import pandas as pd

# Given Mean S/N ratio data (T = 1 Week)

# -----
# -----

# Data for S/N Ratios (Custom Levels)

# Data for S/N Ratios (Levels + Mean S/N Values)

# -----

# Factor Levels

So_levels = [974.457, 984.300, 994.143]
K_levels = [965, 985, 1005]
r_levels = [0.045, 0.055, 0.065]
sigma_levels = [0.15, 0.20, 0.25]

# Mean S/N Ratios

So_SN = [17.010839, 19.466130, 22.820985]
K_SN = [27.160538, 20.247555, 11.889861]
r_SN = [20.109186, 19.162839, 20.025929]
sigma_SN = [18.065565, 19.430277, 21.802112]

# =====

# TABLE 1: FACTOR LEVELS (3×3 layout)

# =====

factor_table = pd.DataFrame({
    "So": So_levels,
    "K": K_levels,
    "r": r_levels,
    "σ": sigma_levels
}, index=["Level 1", "Level 2", "Level 3"])
```

```

print("\nFactor Levels : 1 Week\n")

print(factor_table.to_string())

# MEAN S/N RATIOS (3x3 layout with  $\Delta$  and Rank)

# =====

sn_table = pd.DataFrame({
    "So": So_SN,
    "K": K_SN,
    "r": r_SN,
    "o": sigma_SN
}, index=["Level 1", "Level 2", "Level 3"])

# Calculate Delta & Rank

delta = sn_table.max() - sn_table.min()

rank = delta.rank(ascending=False).astype(int)

# Append Delta & Rank rows

sn_table.loc[" $\Delta$  (Delta)"] = delta

sn_table.loc["Rank"] = rank

sn_table.loc["Rank"] = sn_table.loc["Rank"].apply(lambda x: f"{int(x)}") # force integer display

# Display formatted output

print("\nMean S/N Ratios by Factor Level: 1 Week (Larger-the-Better) \n")

print(sn_table.round(4).to_string())

```

For 1 week

	So	K	r	sigma	BS	SN_ratio
0	974.457	965	0.045	0.15	14.203936	23.048174
1	974.457	985	0.055	0.20	6.724775	16.553556
2	974.457	1005	0.065	0.25	3.728544	11.430786
3	984.300	965	0.055	0.25	25.982638	28.293665
4	984.300	985	0.065	0.15	8.420954	18.507226
5	984.300	1005	0.045	0.20	3.800800	11.597501
6	994.143	965	0.065	0.20	32.135770	30.139774
7	994.143	985	0.045	0.25	19.235089	25.681884
8	994.143	1005	0.055	0.15	4.286124	12.641295

Best Combination: 1 Week (Highest S/N Ratio):

So = 994.143

K = 965

r = 0.045

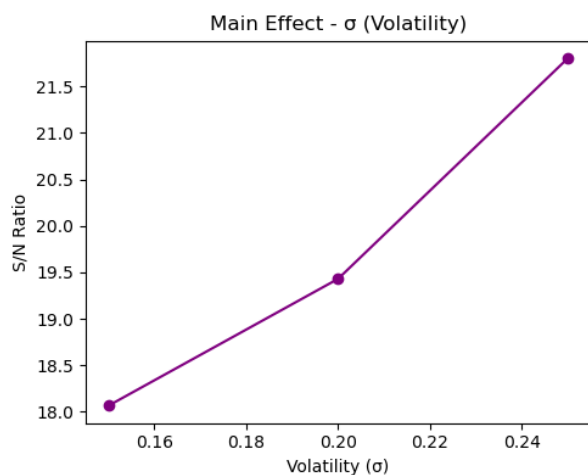
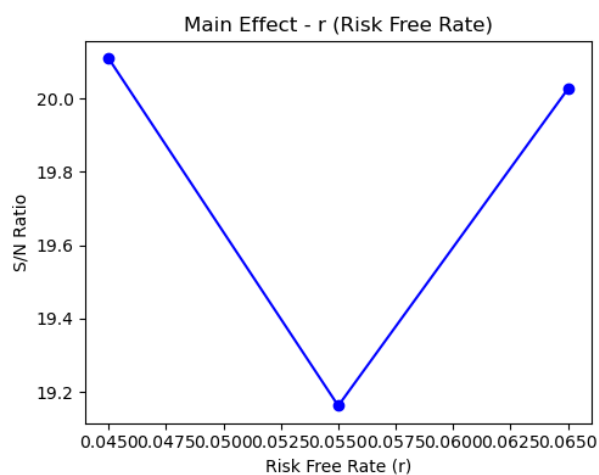
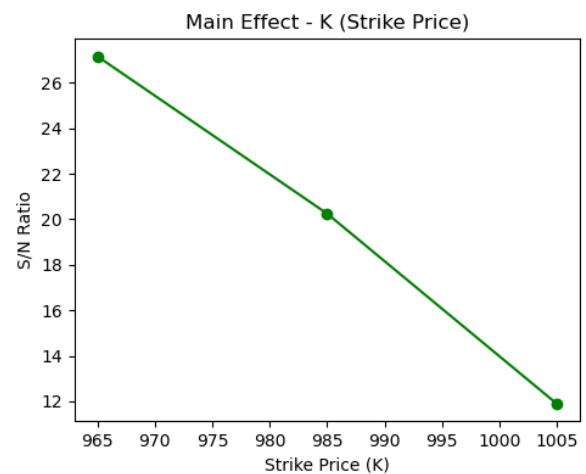
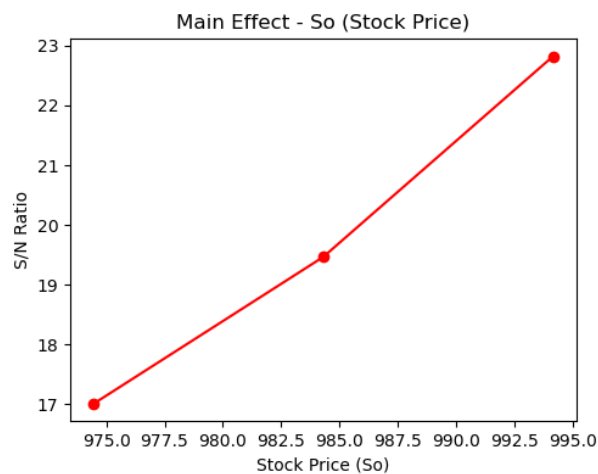
sigma = 0.25

Factor Levels : 1 Week

	So	K	r	σ
Level 1	974.457	965	0.045	0.15
Level 2	984.300	985	0.055	0.20
Level 3	994.143	1005	0.065	0.25

Mean S/N Ratios by Factor Level: 1 Week (Larger-the-Better)

	So	K	r	σ
Level 1	17.010839	27.160538	20.109186	18.065565
Level 2	19.46613	20.247555	19.162839	19.430277
Level 3	22.820985	11.889861	20.025929	21.802112
Δ (Delta)	5.810146	15.270677	0.946347	3.736547
Rank	2	1	4	3



S/N ratio increases with higher Stock Price (S_0) and Volatility (σ), showing improved option stability.

Strike Price (K) displays a strong downward trend, confirming its major inverse impact on option robustness.

Risk Free Rate (r) has a flat pattern, showing minimal sensitivity to option value.

Suggests that K reduces stability, while S_0 and σ improve consistency of option pricing.

For 1 month

Data with S/N Ratio: 1 Month (Larger-the-Better):

	So	K	r	sigma	BS	SN_ratio
0	974.457	965	0.045	0.15	23.920162	27.575282
1	974.457	985	0.055	0.20	19.439162	25.773551
2	974.457	1005	0.065	0.25	17.392079	24.807030
3	984.300	965	0.055	0.25	41.211412	32.300350
4	984.300	985	0.065	0.15	19.218061	25.674191
5	984.300	1005	0.045	0.20	15.215469	23.645707
6	994.143	965	0.065	0.20	43.546009	32.778967
7	994.143	985	0.045	0.25	35.085407	30.902530
8	994.143	1005	0.055	0.15	14.132010	23.004079

Best Combination: 1 Month (Highest S/N Ratio):

So = 994.143

K = 965

r = 0.065

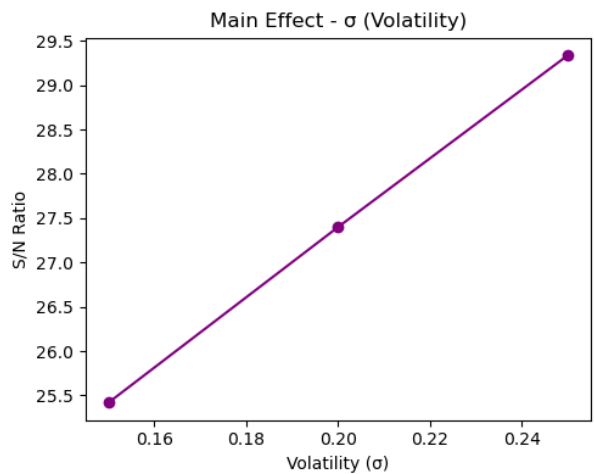
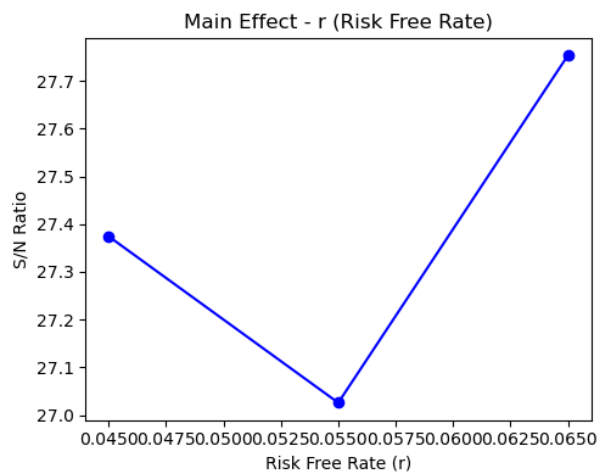
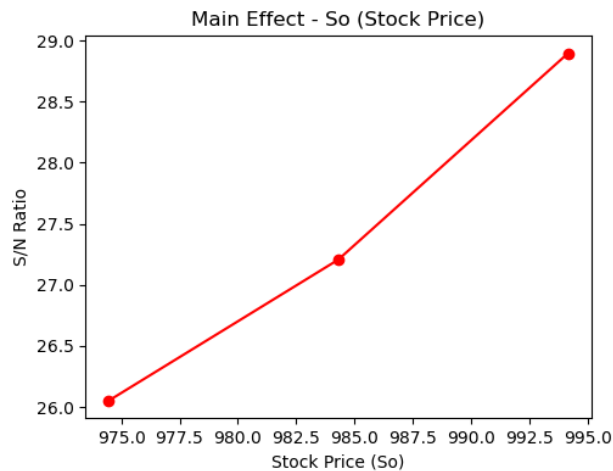
sigma = 0.25

Factor Levels : 1 Month

	So	K	r	σ
Level 1	974.457	965	0.045	0.15
Level 2	984.300	985	0.055	0.20
Level 3	994.143	1005	0.065	0.25

Mean S/N Ratios by Factor Level: 1 Month (Larger-the-Better)

	So	K	r	σ
Level 1	26.051954	30.884866	27.374507	25.417851
Level 2	27.20675	27.450091	27.025993	27.399408
Level 3	28.895192	23.818939	27.753396	29.336637
Δ (Delta)	2.843238	7.065927	0.727403	3.918786
Rank	3	1	4	2



S/N ratio increases with higher Stock Price (So) and Volatility (σ), showing better option stability.

Strike Price (K) shows a clear downward trend, confirming it still has a strong negative impact on robustness.

Risk Free Rate (r) remains almost flat, suggesting minimal influence.

Indicates that K continues to weaken robustness, while So and σ help enhance consistency of the option value.

For 3 months

Data with S/N Ratio: 3 Months (Larger-the-Better):

	So	K	r	sigma	BS	SN_ratio
0	974.457	965	0.045	0.15	39.817817	32.001549
1	974.457	985	0.055	0.20	39.916033	32.022947
2	974.457	1005	0.065	0.25	41.653613	32.393054
3	984.300	965	0.055	0.25	65.784095	36.362418
4	984.300	985	0.065	0.15	37.111176	31.390094
5	984.300	1005	0.045	0.20	34.554314	30.770046
6	994.143	965	0.065	0.20	64.748447	36.224587
7	994.143	985	0.045	0.25	59.365346	35.470660
8	994.143	1005	0.055	0.15	30.853934	29.786211

Best Combination: 3 Months (Highest S/N Ratio):

So = 994.143

K = 965

r = 0.065

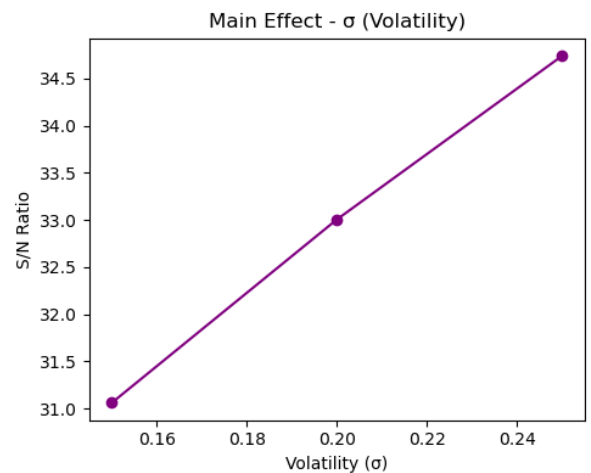
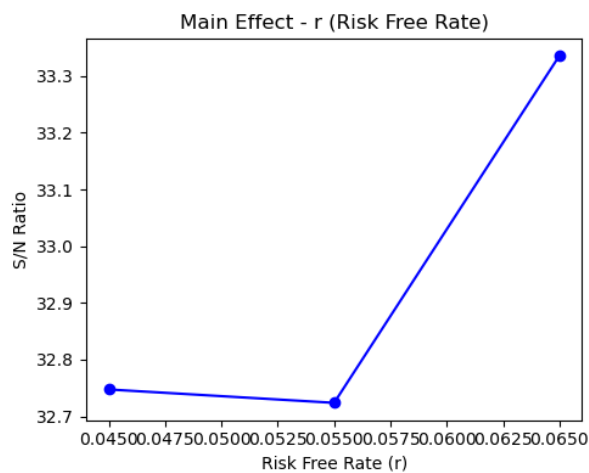
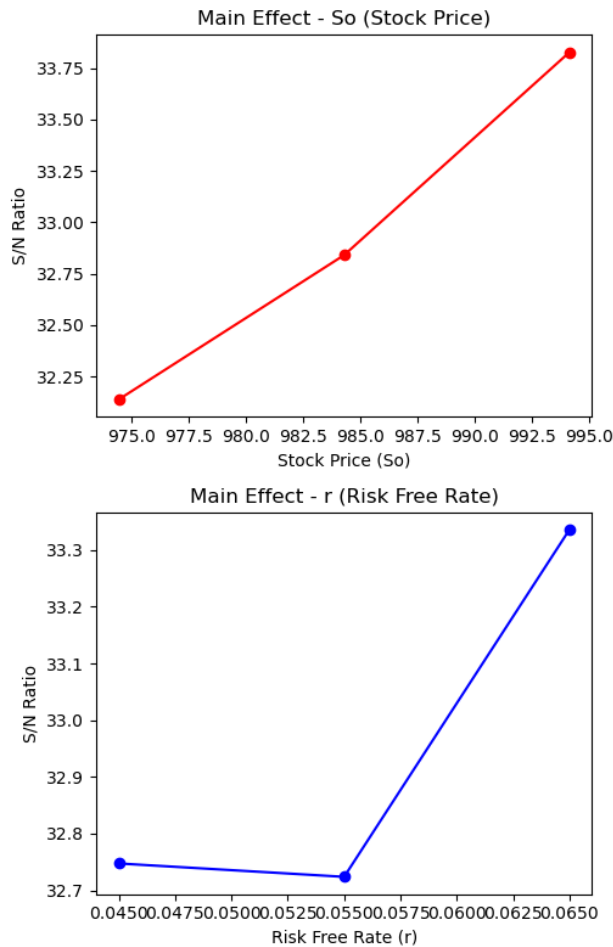
sigma = 0.25

Factor Levels : 3 Months

	So	K	r	σ
Level 1	974.457	965	0.045	0.15
Level 2	984.300	985	0.055	0.20
Level 3	994.143	1005	0.065	0.25

Mean S/N Ratios by Factor Level: 3 Months (Larger-the-Better)

	So	K	r	σ
Level 1	32.139183	34.862851	32.747418	31.059285
Level 2	32.840853	32.961234	32.723859	33.00586
Level 3	33.827153	30.983103	33.335912	34.742044
Δ (Delta)	1.68797	3.879748	0.612053	3.682759
Rank	3	1	4	2



S/N ratio increases with higher Stock Price (S_o) and Volatility (σ), indicating improved option stability at longer maturities.

Strike Price (K) shows a steady downward trend, maintaining its strong inverse relationship with option robustness.

Risk Free Rate (r) continues to have a flat effect, implying negligible influence on option behavior.

Suggests that S_o and σ enhance stability, while K remains the main factor reducing robustness

For optimal combination proof

```
def black_scholes_call(S, K, r, sigma, T):
    d1 = (np.log(S / K) + (r + 0.5 * sigma**2) * T) / (sigma * np.sqrt(T))
    d2 = d1 - sigma * np.sqrt(T)
    call_price = S * norm.cdf(d1) - K * np.exp(-r * T) * norm.cdf(d2)
    return call_price
```

```
print(black_scholes_call(994.143,965,0.065,0.25,30/365))
```

```
48.35323641018567
```

	So	K	r	sigma	BS_Call_Price
1	974.46	965.00	0.045	0.15	23.92
2	974.46	985.00	0.055	0.20	19.44
3	974.46	1005.00	0.065	0.25	17.39
4	984.30	965.00	0.055	0.25	41.21
5	984.30	985.00	0.065	0.15	19.22
6	984.30	1005.00	0.045	0.20	15.22
7	994.14	965.00	0.065	0.20	43.55
8	994.14	985.00	0.045	0.25	35.09
9	994.14	1005.00	0.055	0.15	14.13

By applying the optimal factor combination for 1 month in the Black–Scholes model, we obtain the highest option value (48.35) among all nine L9 experimental runs, confirming that the Taguchi parameters maximize the option's price and robustness.

Conclusion

- Taguchi Method Effectiveness:
The Taguchi approach successfully identified the optimal parameter combination that maximizes option price and model robustness using the Signal-to-Noise ratio.
- Dominance of Strike Price (K):
Across all time periods (1 week, 1 month, and 3 months), the Strike Price consistently ranked as the most influential factor affecting the option value.
- Volatility's Growing Influence:
Volatility (σ) showed an increasing impact over longer time horizons, indicating its major role in determining option price stability.
- Stock Price (So) Consistency:
Stock Price maintained a strong but stable contribution throughout, reinforcing its steady influence on the call option value.
- Risk Free Rate (r) Least Impactful:
The Interest Rate exhibited minimal variation in S/N ratios, suggesting it has the least effect on short-term option pricing behavior.
- Optimal Combination Validation:
The optimal levels derived from the Taguchi analysis produced the highest Black–Scholes value. The best parameter combinations identified were — $So_3-K_1-r_1-\sigma_3$ for the 1-week period, $So_3-K_1-r_3-\sigma_3$ for the 1-month period, and $So_3-K_1-r_3-\sigma_3$ for the 3-month period

Optimal Combination we get after applying Taguchi

For 1 week

- **Stock Price (So) = 994.143 → Level 3**
- **Strike Price (K) = 965 → Level 1**
- **Risk Free Rate (r) = 0.045 → Level 1**
- **Volatility (σ) = 0.25 → Level**

For 1 month

- **Stock Price (So) = 994.143 → Level 3**
- **Strike Price (K) = 965 → Level 1**
- **Risk Free Rate (r) = 0.065 → Level 3**
- **Volatility (σ) = 0.25 → Level 3**

For 3 months

- **Stock Price (S_0) = 994.143 → Level 3**
- **Strike Price (K) = 965 → Level 1**
- **Risk Free Rate (r) = 0.065 → Level 3**
- **Volatility (σ) = 0.25 → Level 3**

Discussion

This study presents a compelling application of the Taguchi L9 orthogonal array method to the Black-Scholes Model (BSM) for call option pricing, specifically within a Multiple -period framework. By strategically reducing the number of experimental runs from 81 (full factorial design) to just 9, the authors demonstrate how statistical design of experiments (DOE) can be used to efficiently explore the influence of multiple financial parameters without compromising analytical depth.

The four parameters under investigation—underlying asset price (S_0), strike price (K), risk-free interest rate (r), and volatility (σ)—were each tested at three levels. Using the Black-Scholes formula, the authors calculated the call option value for each experimental combination and then applied statistical tools such as ANOM (Analysis of Means), S/N ratio (Signal-to-Noise), and ANOVA (Analysis of Variance) to interpret the results.

The findings consistently showed that the underlying asset price (S_0) had the most significant impact on the call option value, contributing over 81% to the variation according to ANOVA. Strike price (K) followed with a 17% contribution, while interest rate and volatility had minimal effects in this single-period context. These results were reinforced by the S/N ratio and Tukey method, which confirmed the statistical significance of S_0 over the other parameters.

The regression model derived from the experimental data exhibited an excellent fit (R^2), indicating that the chosen parameters effectively explained the variation in call option values. Residual plots further validated the model's reliability, showing normal distribution and absence of outliers.

Importantly, the study identified an optimal parameter combination S_0, K, r, σ which yielded the highest call option value. This insight is particularly valuable for financial analysts and traders seeking to understand how sensitive option pricing is to changes in market inputs.

Overall, the paper highlights the power of the Taguchi method in financial modeling, offering a structured, cost-effective approach to parameter analysis. It bridges the gap between statistical engineering techniques and financial economics, paving the way for future research in multi-period models, risk assessment, and derivative pricing optimization.

Future scope

- Integration with Machine Learning.

Machine learning (ML) techniques can examine past financial data to predict asset prices, volatility, and market behaviors. - Pattern Recognition: ML is particularly adept at uncovering intricate, non-linear connections within data, aiding in the identification of anomalies, arbitrage chances, or early indicators of market changes. Portfolio Optimization: ML has the capability to adjust portfolio distributions dynamically in response to shifting market conditions and investor interests. - Risk Assessment: Algorithms are able to evaluate credit risk, counterparty risk, and systemic risk more effectively than conventional models.

- Sensitivity Analysis Across Market Conditions

Stress Testing: Analyzing how financial models react to extreme situations such as market crashes, interest rate increases, or geopolitical disturbances. Scenario Analysis: Assessing model results under varying assumptions (e.g., inflation levels, GDP growth) to highlight weaknesses. Parameter Impact: Determining which inputs (e.g., volatility, correlation) significantly affect derivative pricing or the value of a portfolio. Robustness Checks: Confirming that models maintain reliability across diverse market conditions, thereby minimizing the risk of overfitting.

- Application to Other Derivatives

Options: Broadening models to evaluate exotic options (e.g., barrier, Asian, lookback) through advanced numerical approaches or ML. Swaps and Forwards: Utilizing valuation methods for interest rate swaps, currency forwards, and commodity derivatives. Credit Derivatives: Analyzing credit default swaps (CDS) and collateralized debt obligations (CDOs) with improved risk metrics. -Structured Products: Assisting in the creation and valuation of intricate instruments like equity-linked notes or basket options.

- Real-Time Decision Support Systems

Live Market Feeds: Merging real-time data streams to immediately refresh models and dashboards. Automated Alerts: Issuing notifications for risk violations, arbitrage prospects, or notable market fluctuations. Interactive Dashboards: Equipping traders and analysts with user-friendly interfaces to simulate trades and evaluate results. AI-Driven Recommendations: Providing actionable insights based on prevailing market conditions and historical data patterns.

- Educational Use in Finance Curricula

Hands-On Learning: Enabling students to construct and test models using actual data and tools such as Python, R, or MATLAB. Case Studies: Integrating historical financial crises or market incidents to illustrate risk management and model limitations. Simulation Platforms: Offering virtual trading settings where students can link theory

to practice. Interdisciplinary Training: Connecting finance with data science, statistics, and computer science to equip students for contemporary financial careers.

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<https://www.nseindia.com/market-data/live-market-indices>

Thank you